

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

# Chapter 14: Probability

Every formula and result of the chapter on one sheet

|    |                                  |                                                                                       |
|----|----------------------------------|---------------------------------------------------------------------------------------|
| 1  | Probability of an event          | $P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ |
| 2  | Range of probability             | $0 \leq P(E) \leq 1$                                                                  |
| 3  | Impossible event                 | $P(E) = 0$                                                                            |
| 4  | Sure event                       | $P(E) = 1$                                                                            |
| 5  | Complementary event              | $P(\bar{E}) = 1 - P(E)$                                                               |
| 6  | All elementary events            | The sum of their probabilities is 1                                                   |
| 7  | At least one                     | $P(\text{at least one}) = 1 - P(\text{none})$                                         |
| 8  | One coin, two coins, three coins | 2, 4 and 8 outcomes                                                                   |
| 9  | One die, two dice                | 6 and 36 outcomes                                                                     |
| 10 | Pack of cards                    | 52 cards: 26 red, 26 black; 4 suits of 13                                             |
| 11 | Face cards                       | 12 (jack, queen, king in each suit)                                                   |
| 12 | Days in a year                   | Ordinary year: 52 weeks + 1 day; leap year: 52 weeks + 2 days                         |